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TRƯỜNG ĐẠI HỌC KHOA HỌC TỰ NHIÊN, ĐHQG - HCM
KHOA CÔNG NGHỆ THÔNG TIN

APPLICATIONS OF NATURAL LANGUAGE PROCESSING IN INDUSTRY

An Overview

TS. Nguyễn Hồng Bửu Long

Overview

Content: 11 weeks

- Introduction
- Development Infrastructure & Tooling
- Data Management
- Model & Parameter Optimization
- Deployment
- Agentic AI
- Continual Learning
- Data Privacy & Model Robustness
- Teams and Project Management
- Ethics
- Case Study: Data Annotation & Demo Applications

Assessments:

- Lab Assignments (30%)
- Group Projects:
 - Seminar: 30%
 - Final Evaluation: 40%
- Bonus marks: Q&A, Discussion...

1. Core NLP Tasks

Core NLP Tasks

Dan Jurafsky



Language Technology

making good progress

mostly solved

Spam detection

Let's go to Agra! ✓

Buy V1AGRA ... ✗

Part-of-speech (POS) tagging

ADJ ADJ NOUN VERB ADV

Colorless green ideas sleep furiously.

Named entity recognition (NER)

PERSON ORG LOC

Einstein met with UN officials in Princeton

Sentiment analysis

Best roast chicken in San Francisco! 👍

The waiter ignored us for 20 minutes. 👎

Coreference resolution

Carter told Mubarak he shouldn't run again.

Word sense disambiguation (WSD)

I need new batteries for my *mouse*.

Parsing

I can see Alcatraz from the window!

Machine translation (MT)

第13届上海国际电影节开幕...

The 13th Shanghai International Film Festival...

Information extraction (IE)

You're invited to our dinner party, Friday May 27 at 8:30



still really hard

Question answering (QA)

Q. How effective is ibuprofen in reducing fever in patients with acute febrile illness?

Paraphrase

XYZ acquired ABC yesterday

ABC has been taken over by XYZ

Summarization

The Dow Jones is up

The S&P500 jumped

Housing prices rose

Economy is good

Dialog

Where is Citizen Kane playing in SF?

Castro Theatre at 7:30. Do you want a ticket?



Spam detection

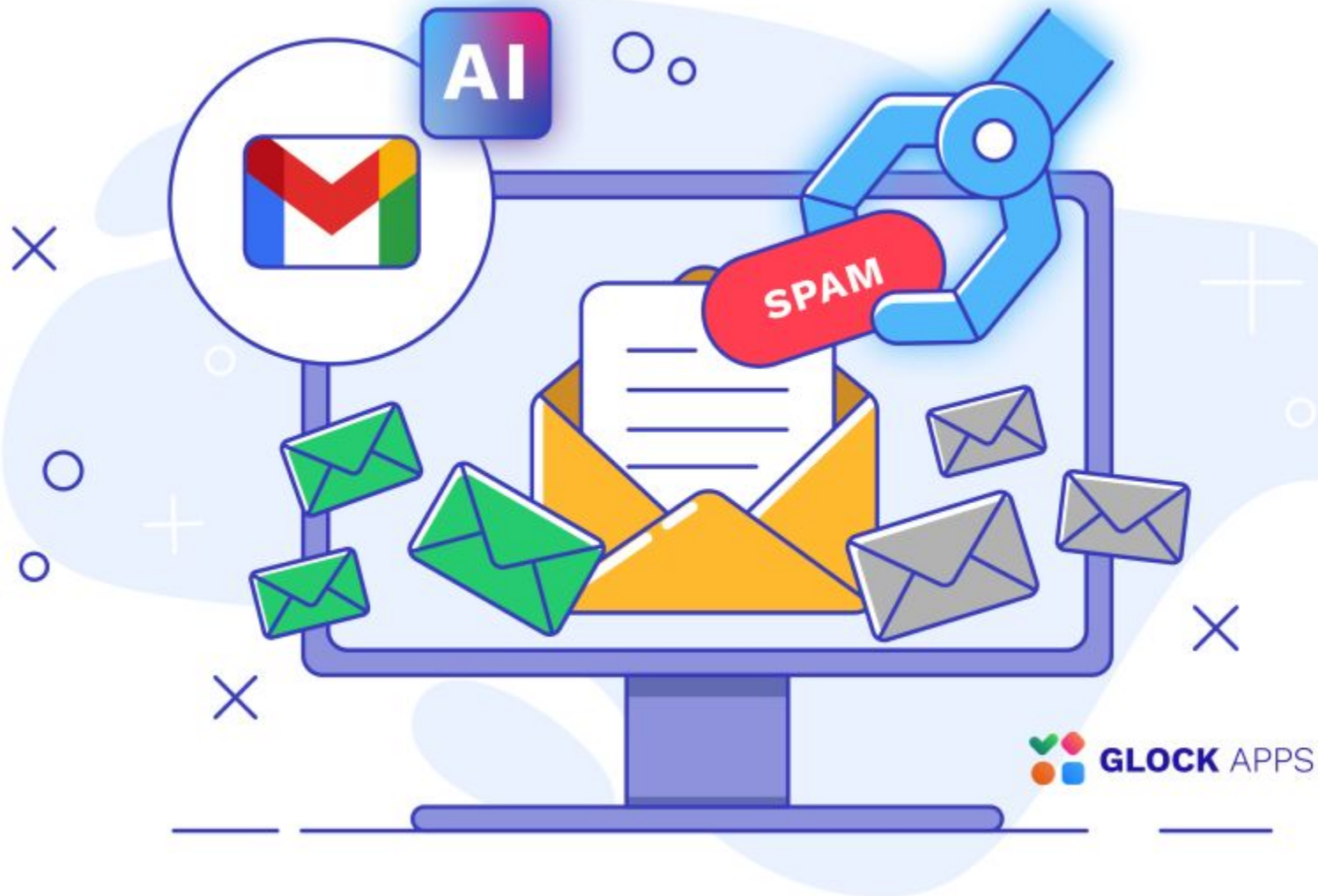


Image: <https://glockapps.com/blog/gmails-biggest-security-update-ai-powered-spam-filter/>

Part-of-speech (POS) tagging

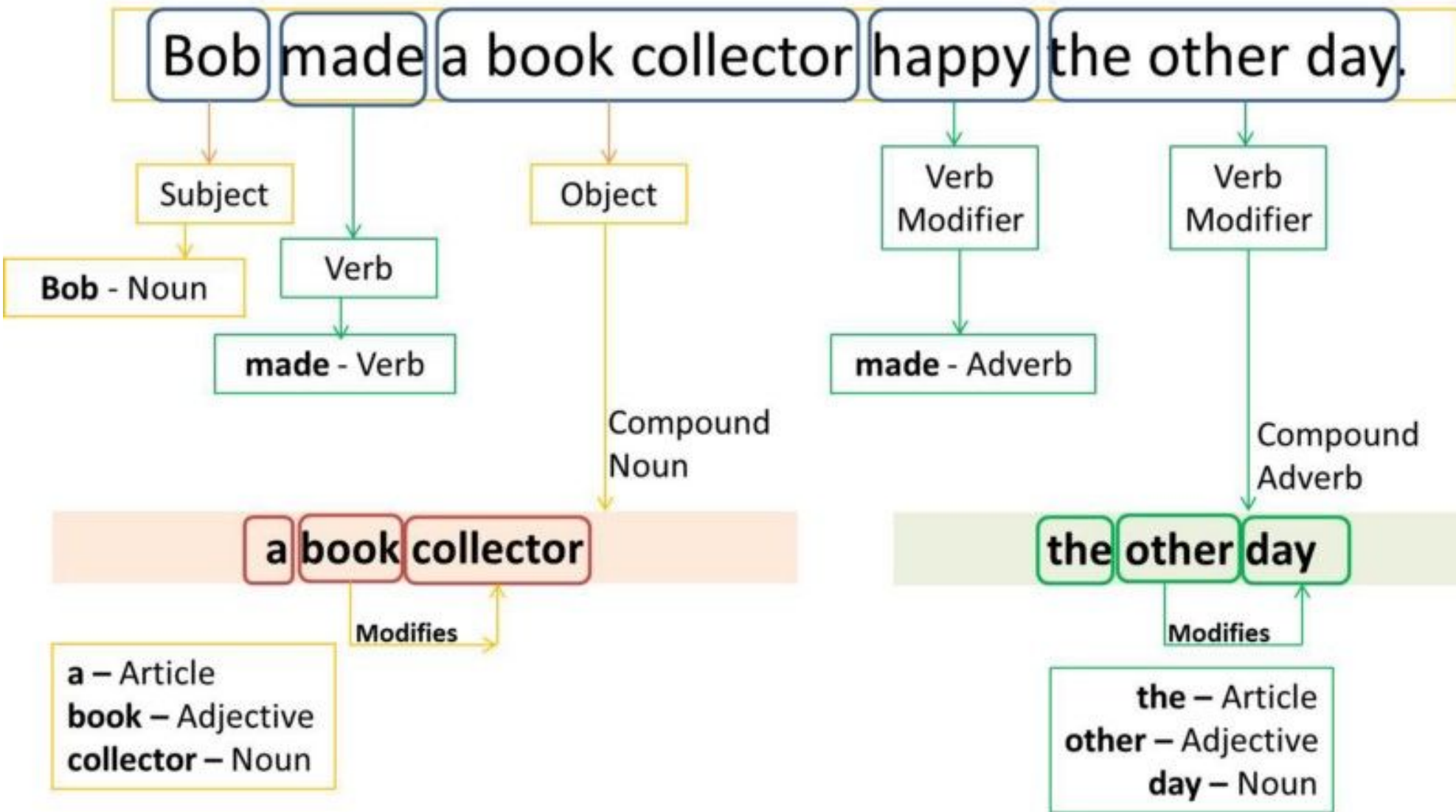


Image: <https://www.freecodecamp.org/news/an-introduction-to-part-of-speech-tagging-and-the-hidden-markov-model-953d45338f24/>

Named Entity Recognition (NER)

In fact, the **Chinese** **NORP** market has the **three** **CARDINAL** most influential names of the retail and tech space – **Alibaba** **GPE** , **Baidu** **ORG** , and **Tencent** **PERSON** (collectively touted as **BAT** **ORG**), and is betting big in the global **AI** **GPE** in retail industry space . The **three** **CARDINAL** giants which are claimed to have a cut-throat competition with the **U.S.** **GPE** (in terms of resources and capital) are positioning themselves to become the ‘future **AI** **PERSON** platforms’. The trio is also expanding in other **Asian** **NORP** countries and investing heavily in the **U.S.** **GPE** based **AI** **GPE** startups to leverage the power of **AI** **GPE** . Backed by such powerful initiatives and presence of these conglomerates, the market in APAC AI is forecast to be the fastest-growing **one** **CARDINAL** , with an anticipated **CAGR** **PERSON** of **45%** **PERCENT** over **2018 - 2024** **DATE** .

To further elaborate on the geographical trends, **North America** **LOC** has procured **more than 50%** **PERCENT** of the global share in **2017** **DATE** and has been leading the regional landscape of **AI** **GPE** in the retail market. The **U.S.** **GPE** has a significant credit in the regional trends with **over 65%** **PERCENT** of investments (including M&As, private equity, and venture capital) in artificial intelligence technology. Additionally, the region is a huge hub for startups in tandem with the presence of tech titans, such as **Google** **ORG** , **IBM** **ORG** , and **Microsoft** **ORG** .

Image: <https://devopedia.org/named-entity-recognition>

Sentiment Analysis



Review



Coreference Resolution

mention

John told **Sally** that **she** should come watch **him** play the **violin**.

antecedent

John told Sally that **she** should come watch him play the violin.


coreferent

John told Sally that **she** should come watch him play the violin.


cluster

John told **Sally** that **she** should come watch **him** play the **violin**.

anaphoric

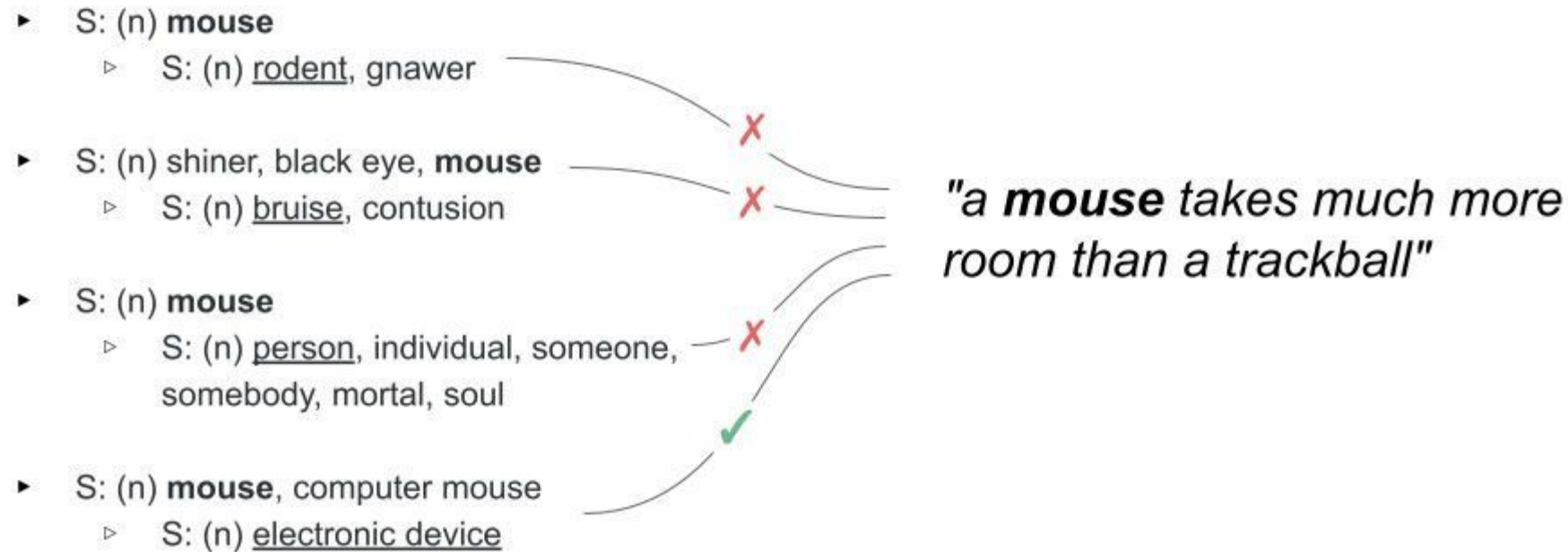
John told Sally that she should come watch him play the violin.


non-anaphoric

John told Sally that she should come watch him play the **violin**.


Image: <https://lumosbusiness.com/how-valuable-is-sentiment-analysis-for-brands/>

Word Sense Disambiguation (WSD)



Parsing

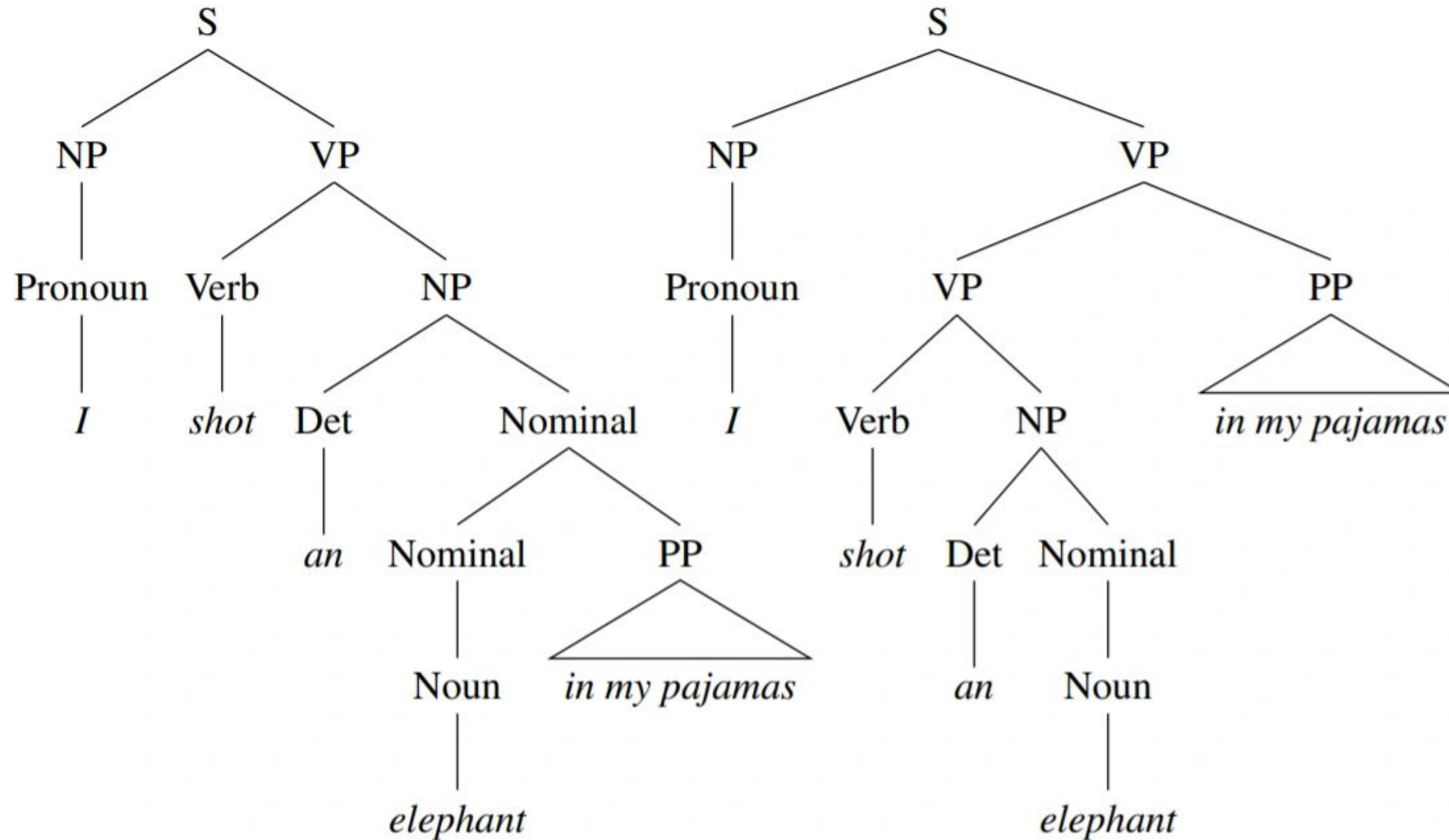


Image: <https://www.analyticsvidhya.com/blog/2021/12/dependency-parsing-in-natural-language-processing-with-examples/>

Machine Translation (MT)

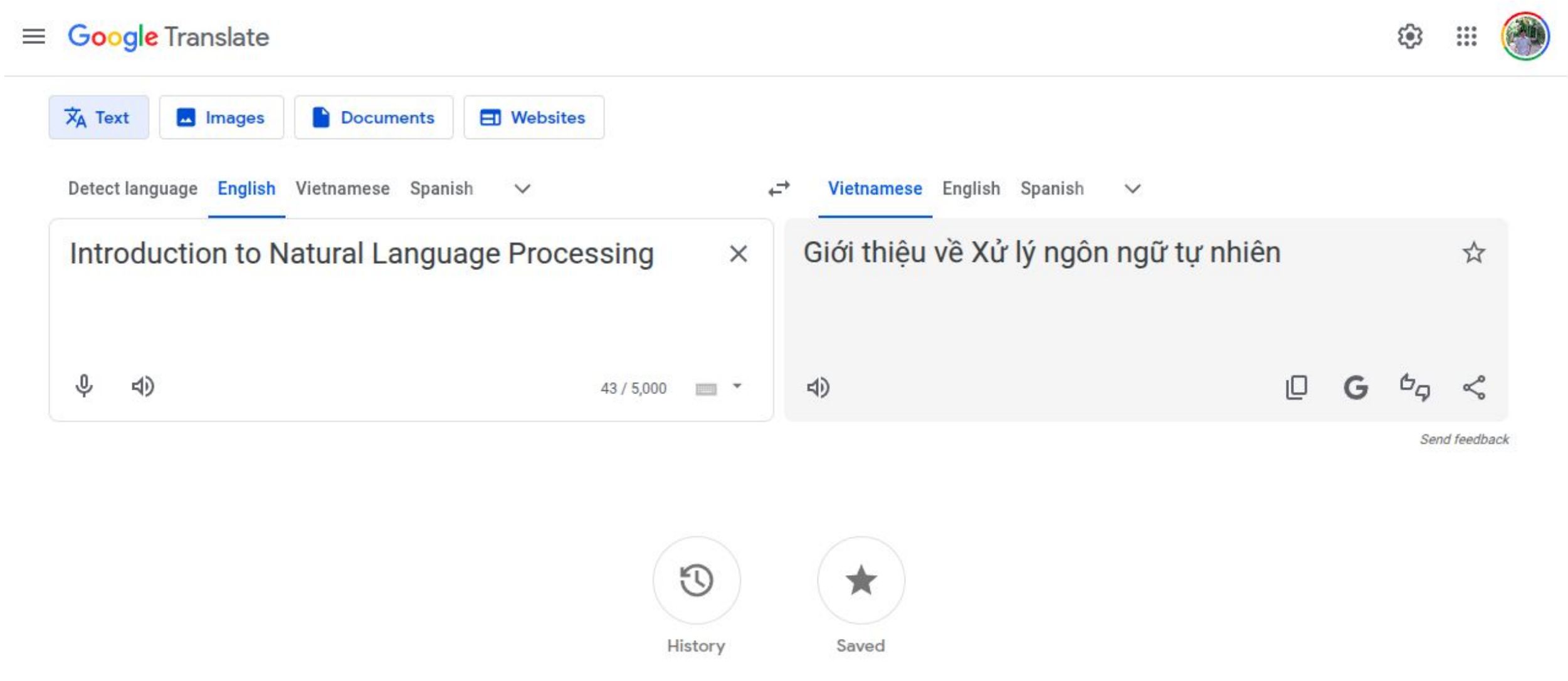
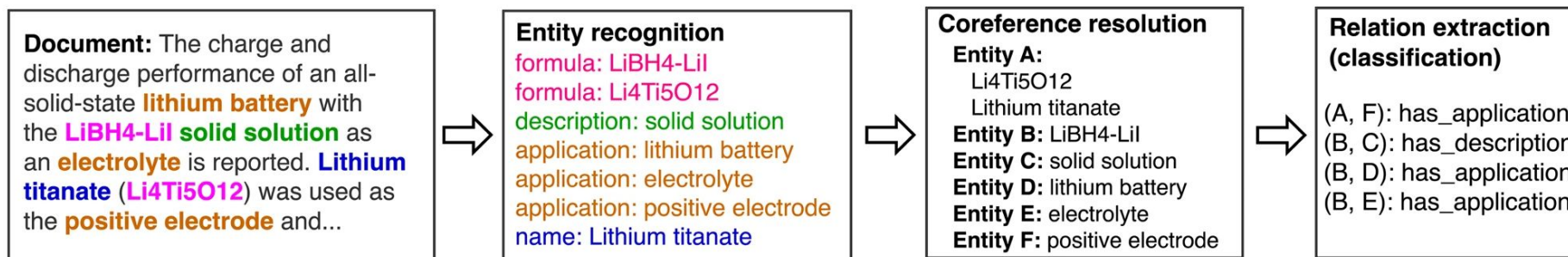


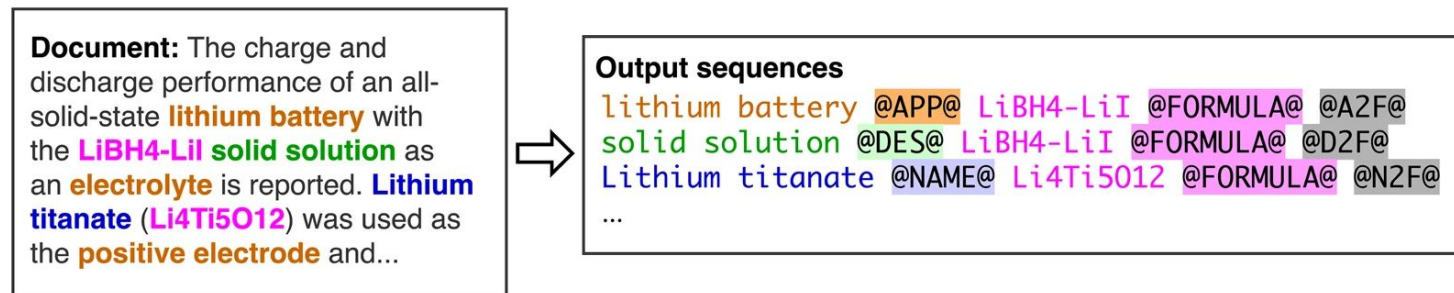
Image: <https://www.analyticsvidhya.com/blog/2021/12/dependency-parsing-in-natural-language-processing-with-examples/>

Information Extraction (IE)

(a) **Previous:** Multi-step (pipeline) named entity recognition and relationship extraction



(b) **Previous:** seq2seq enumerates relationships as 2-tuples



(c) **This work:** Hierarchical entity relationships without explicit enumeration

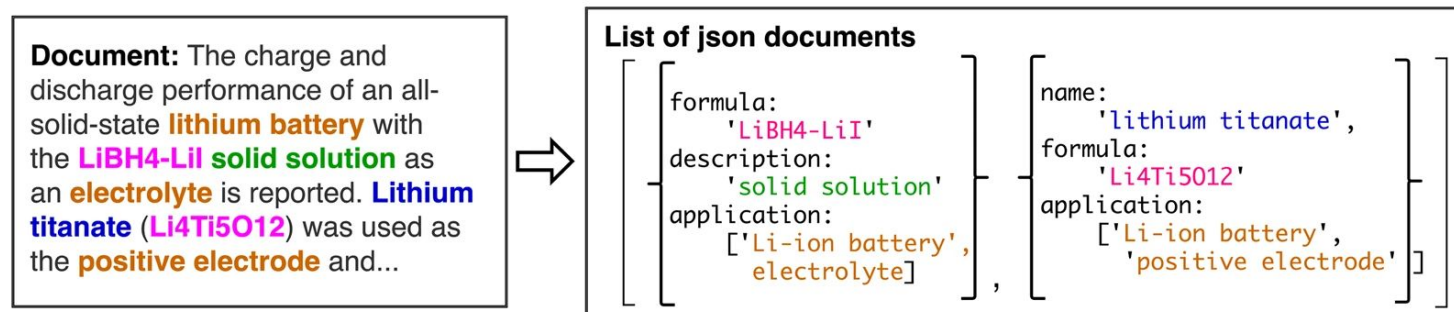


Image: <https://www.nature.com/articles/s41467-024-45563-x/figures/1>

Question Answering (QA)

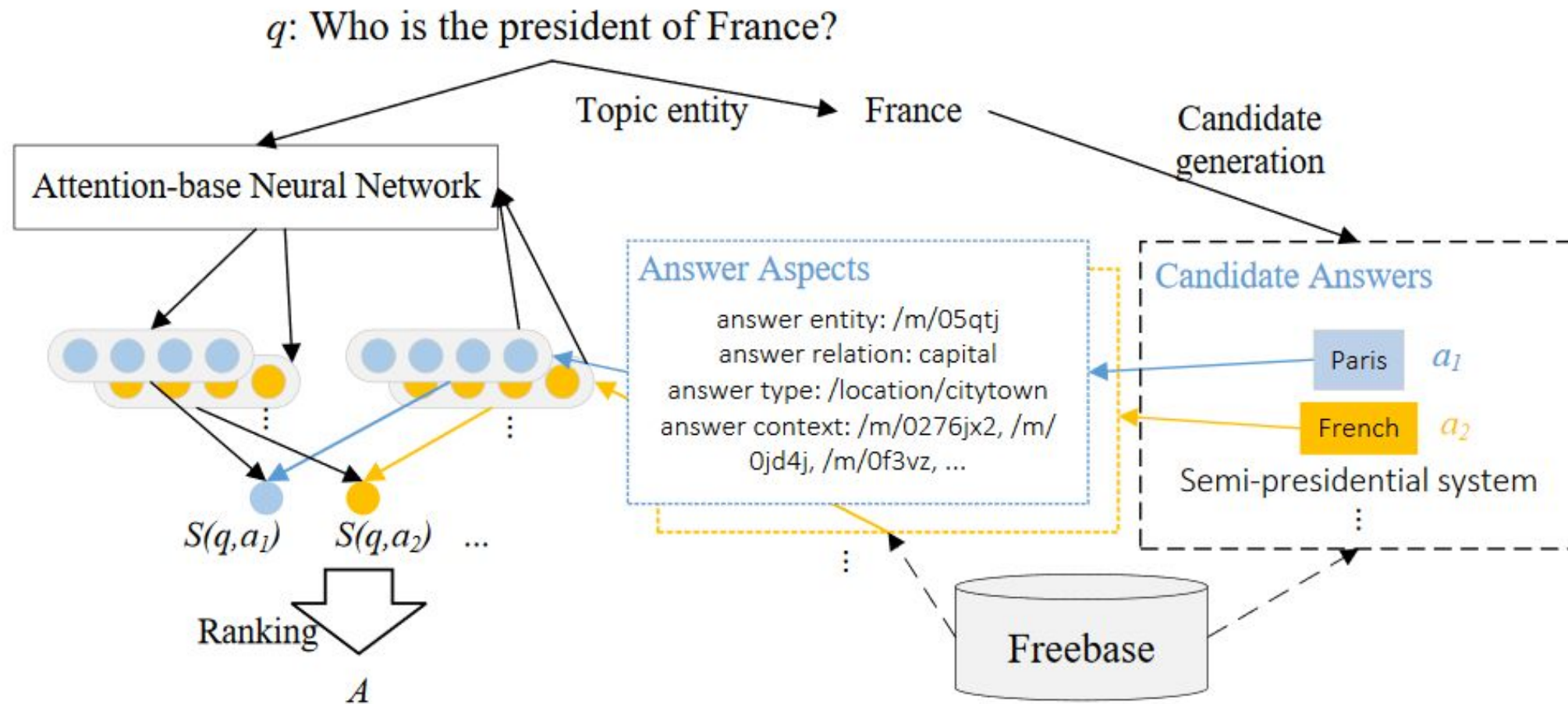


Image: <https://devopedia.org/question-answering>

Paraphrase

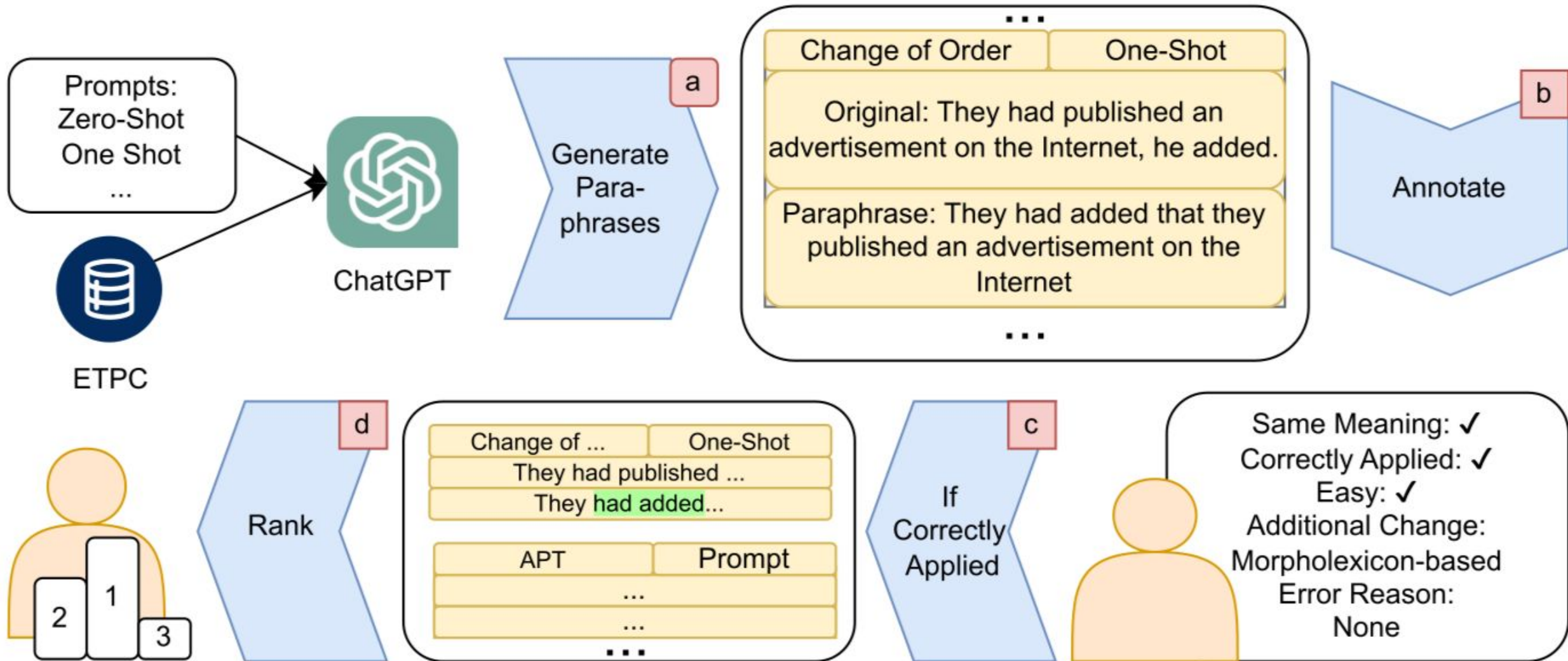
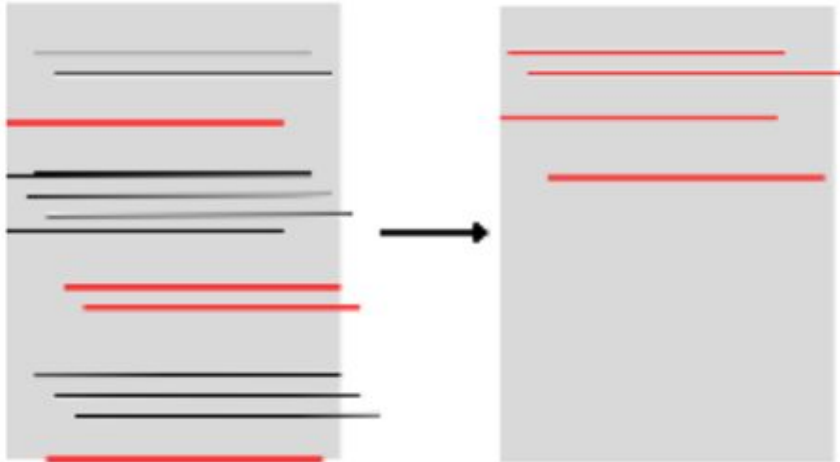


Image: <https://devopedia.org/question-answering>

Summarization

NLP Text Summarization

Extractive



Abstractive

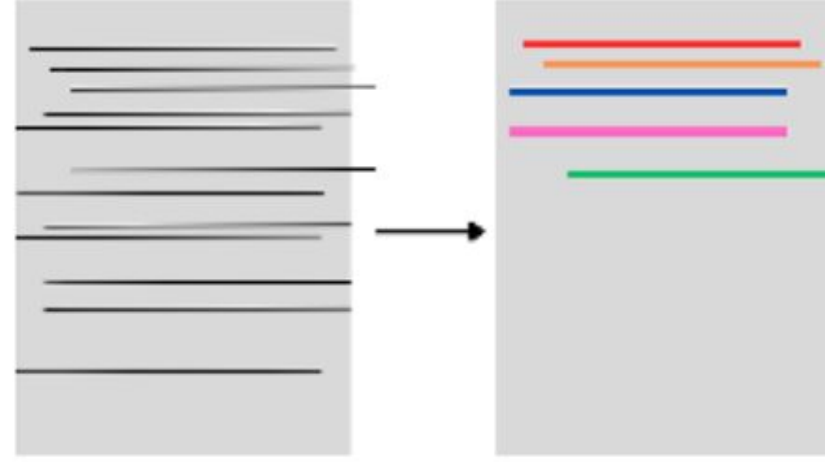


Image: <https://devopedia.org/question-answering>

Dialog/Chatbot/Conversation

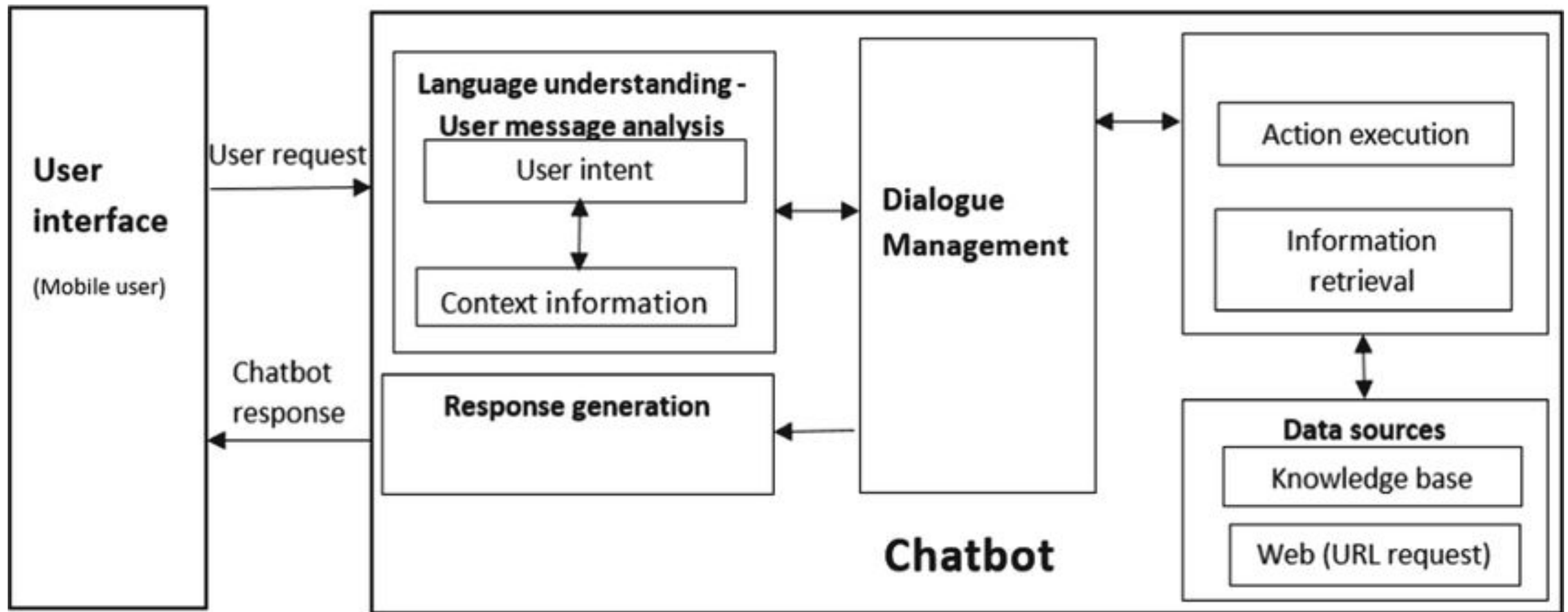
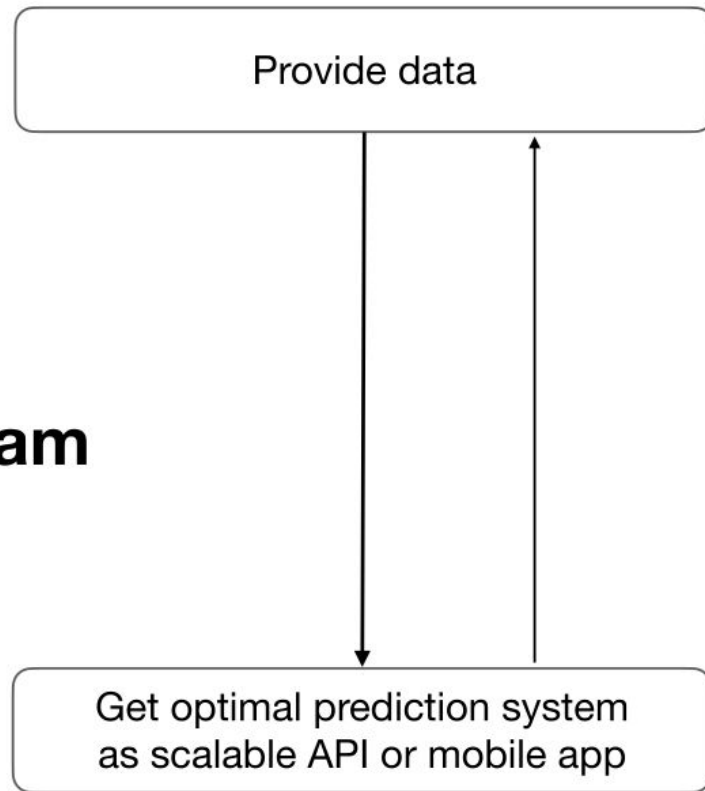


Image: <https://devopedia.org/question-answering>

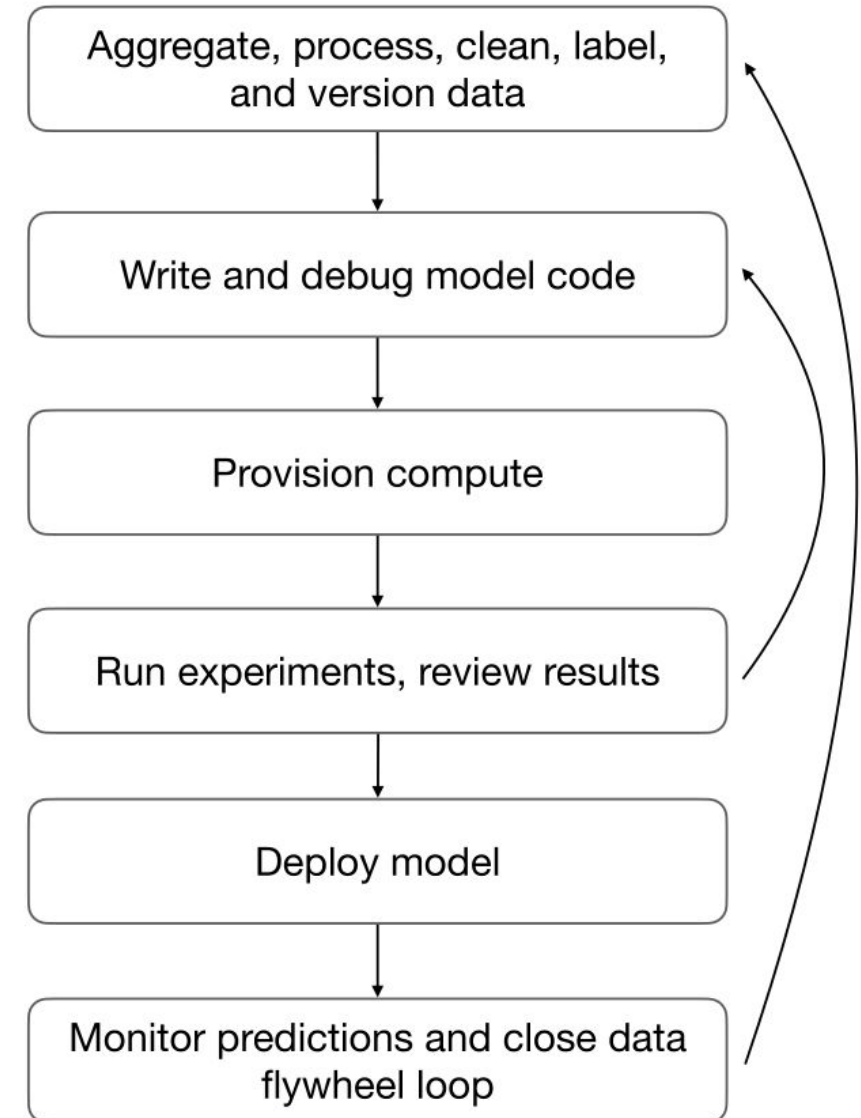
2. Development Infrastructure & Tooling

Development Infrastructure & Tooling

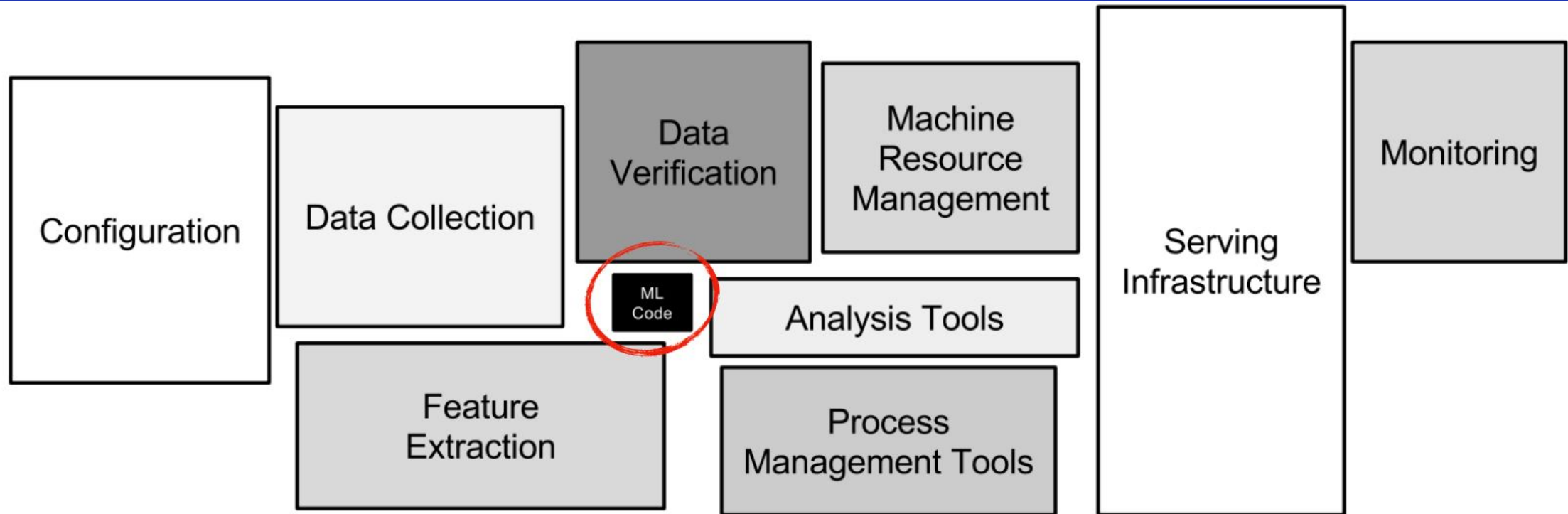
Dream



Reality



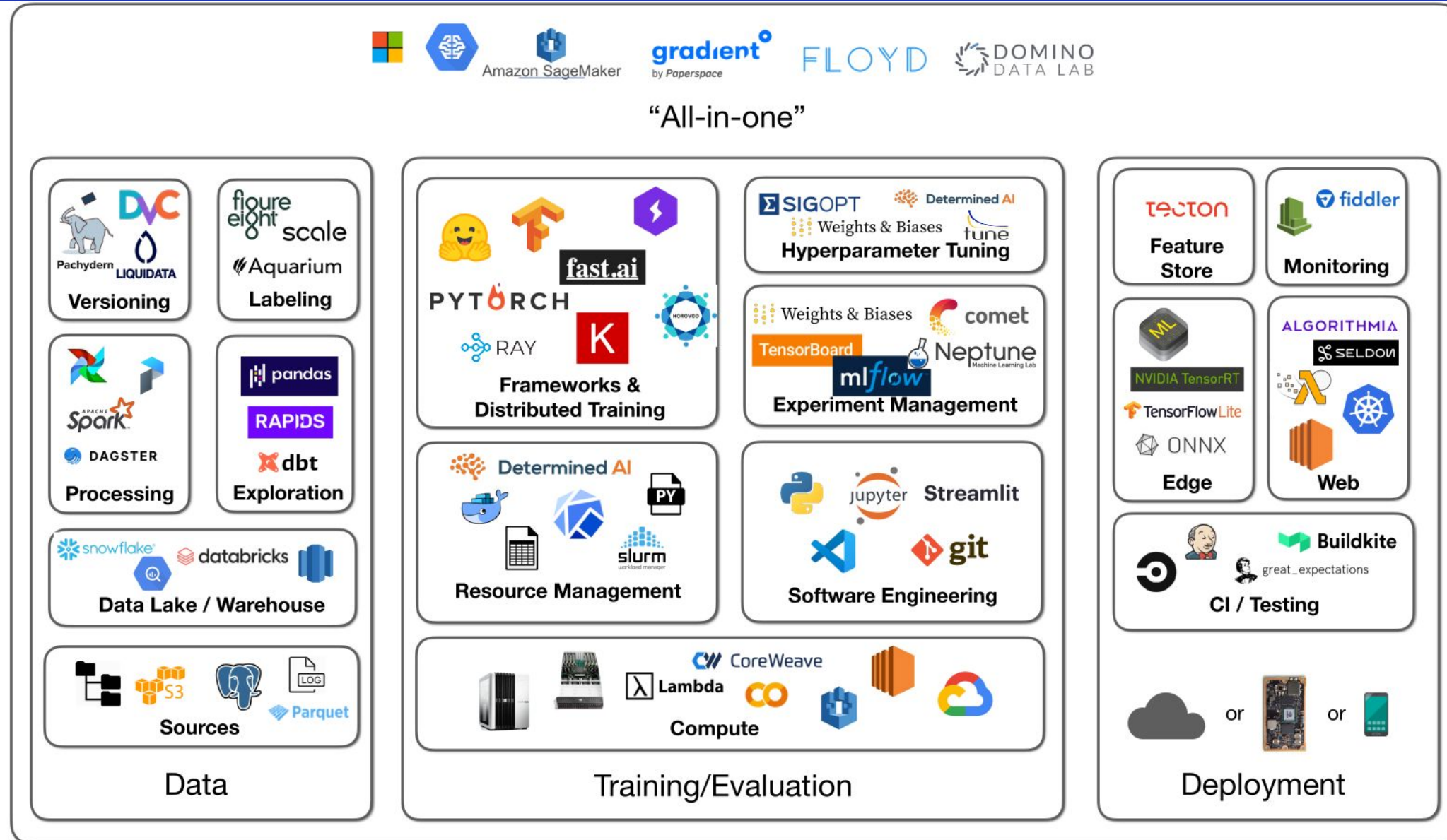
Development Infrastructure & Tooling



Machine Learning: The High-Interest Credit Card of Technical Debt

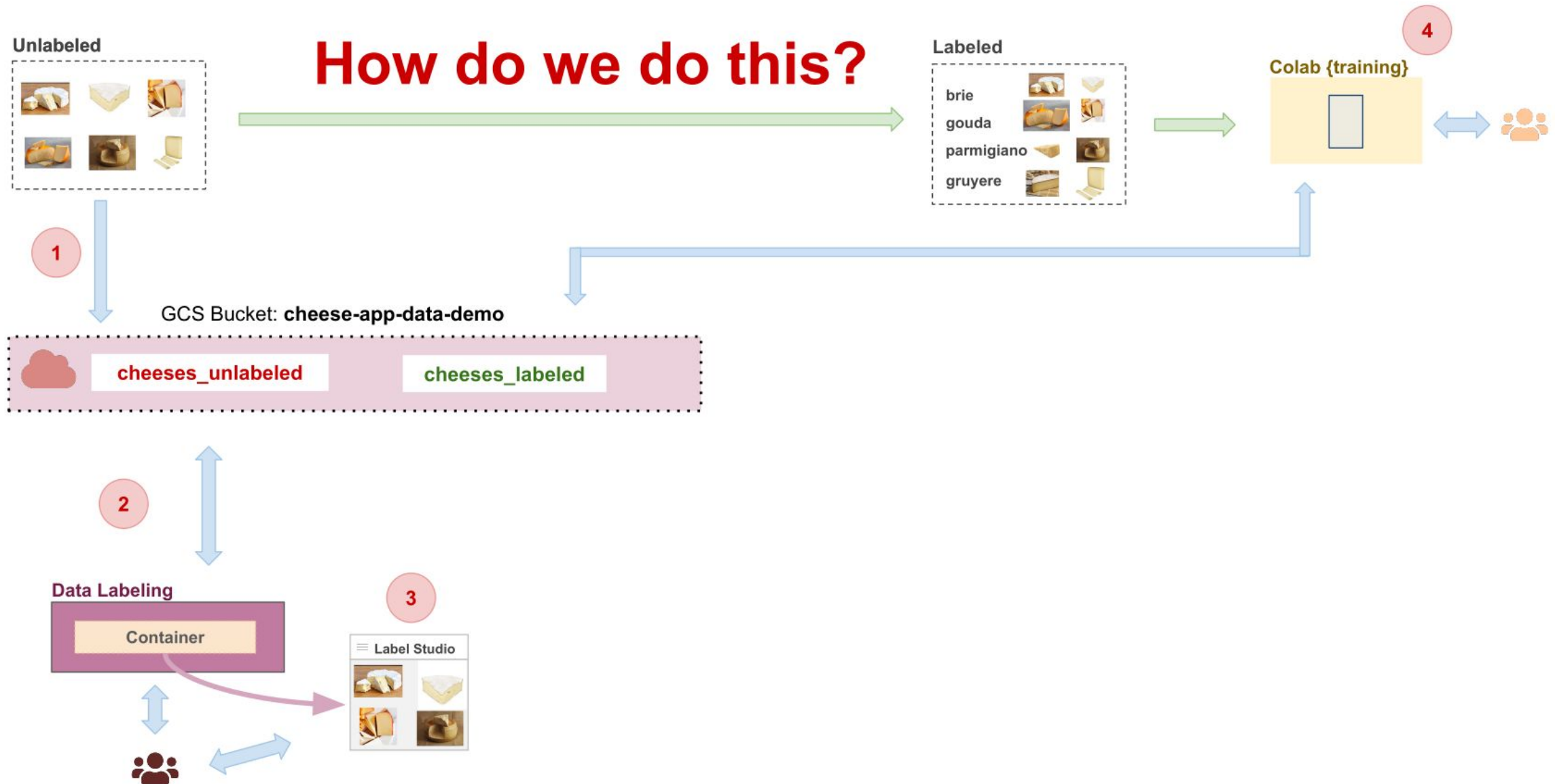
D. Sculley, Gary Holt, Daniel Golovin, Eugene Davydov,
Todd Phillips, Dietmar Ebner, Vinay Chaudhary, Michael Young

Development Infrastructure & Tooling

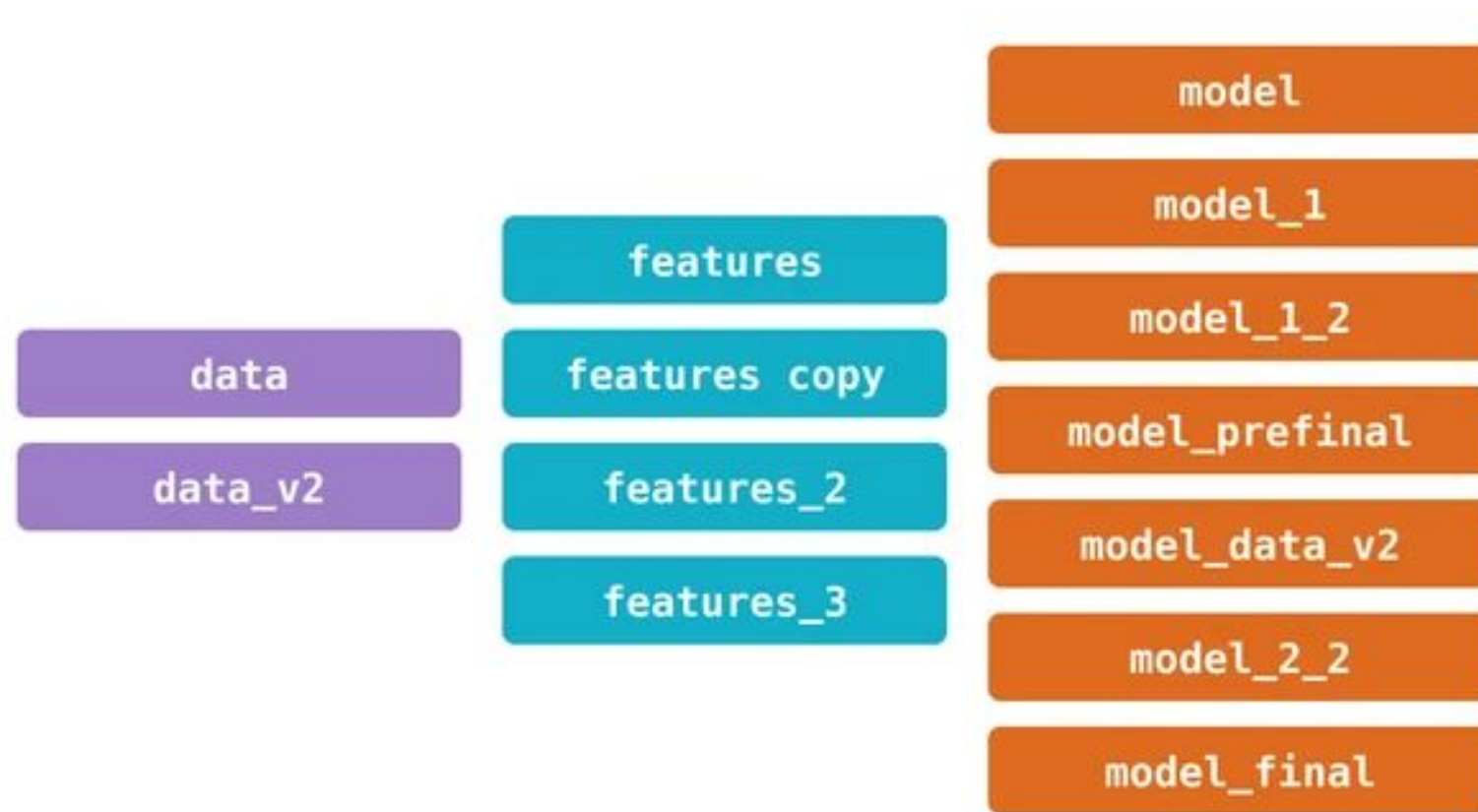


3. Data Management

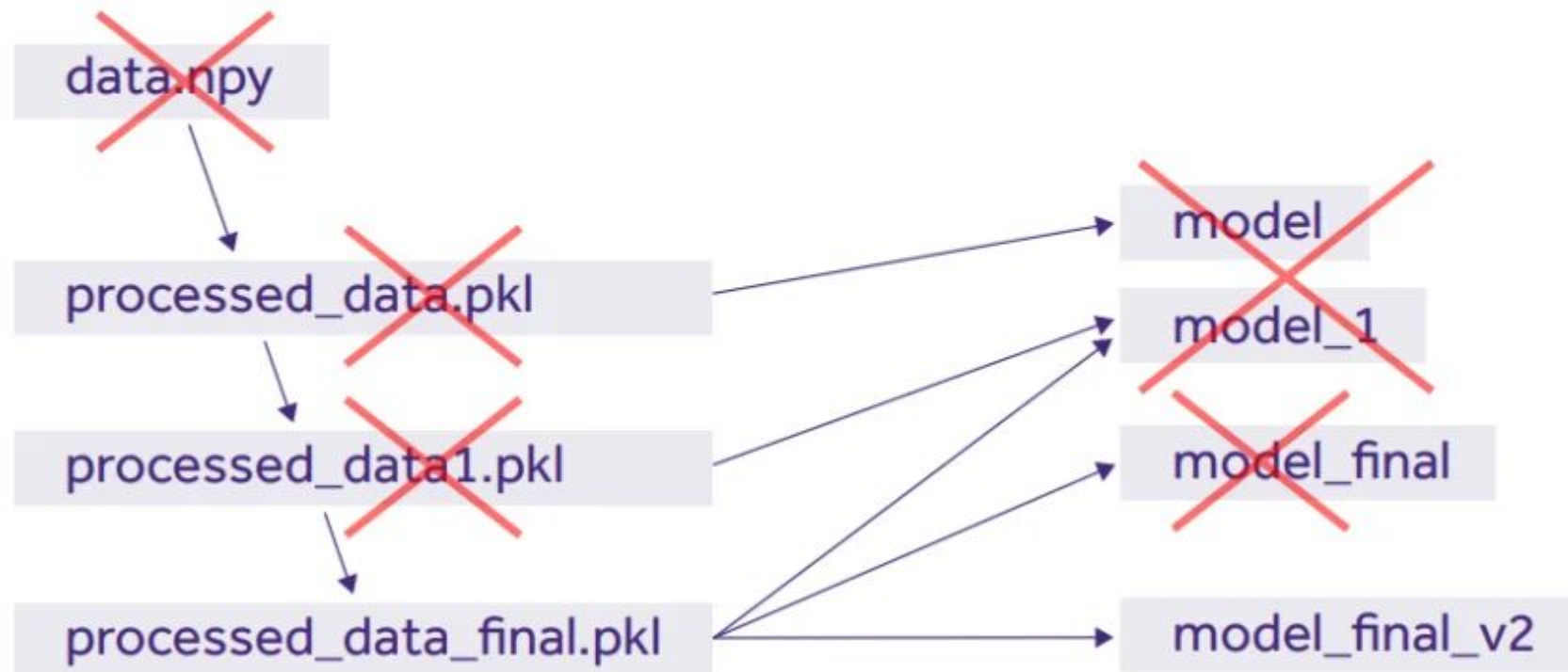
Data Management



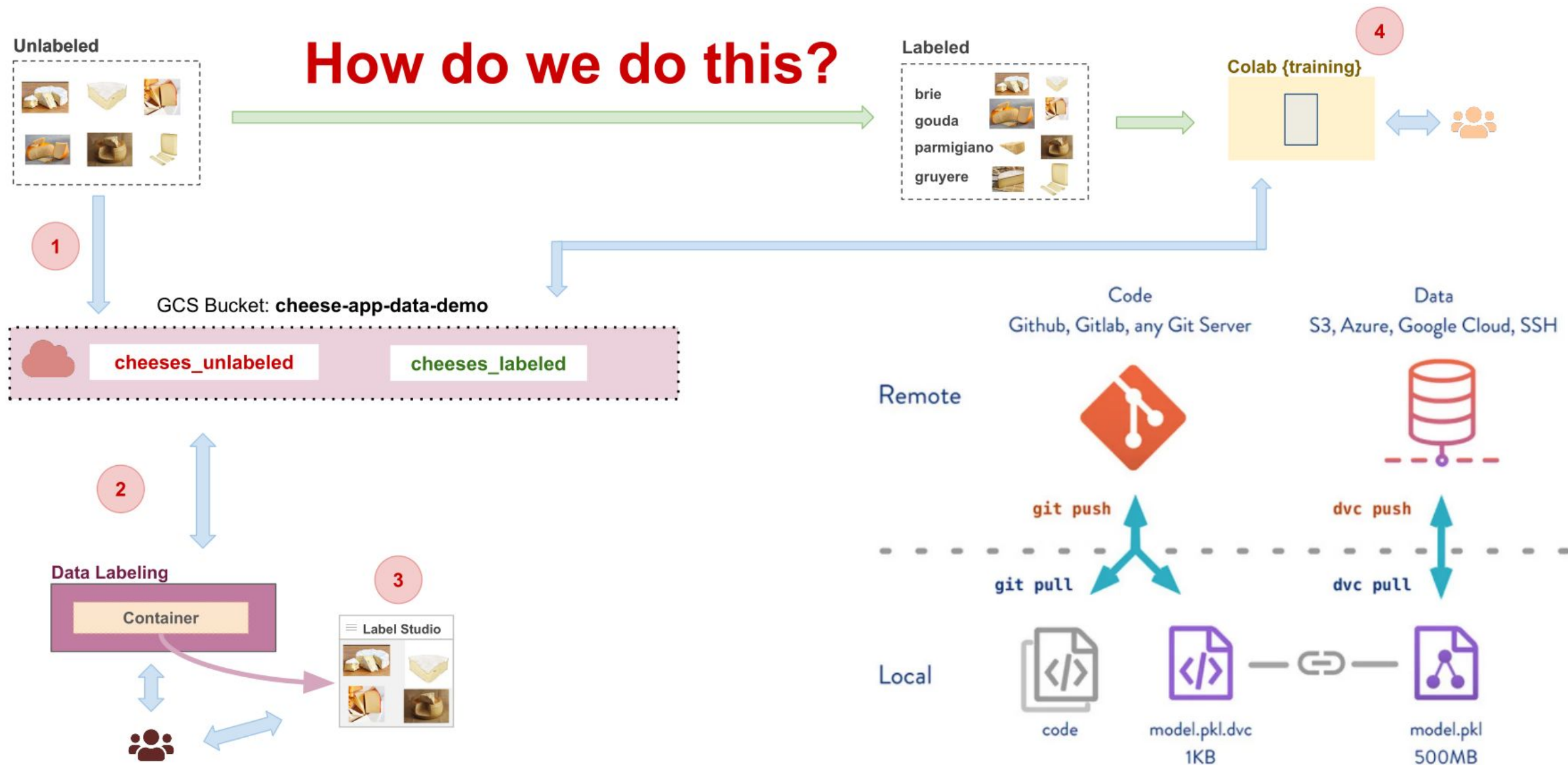
Data Management



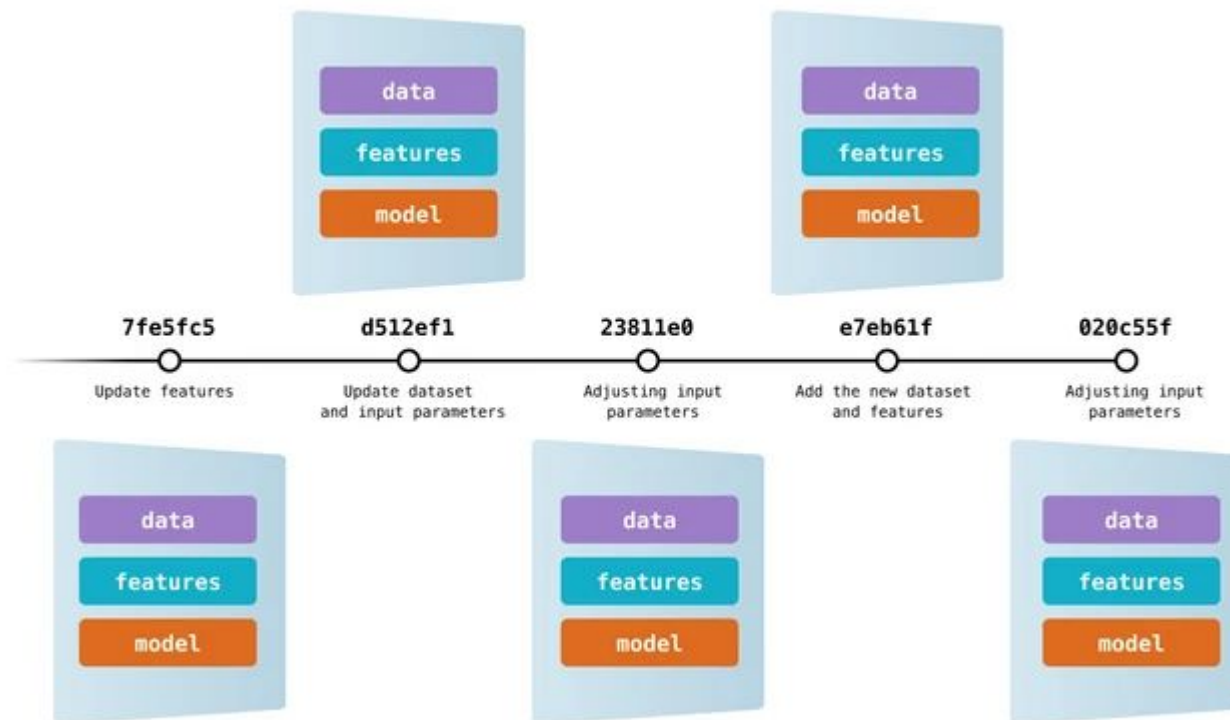
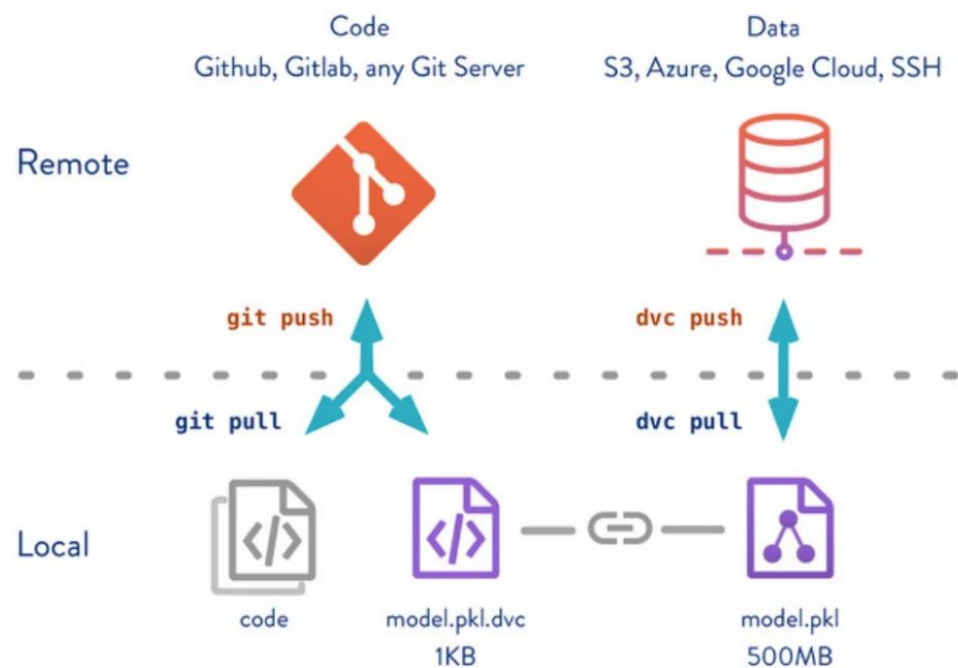
Data Version Control



Data Management



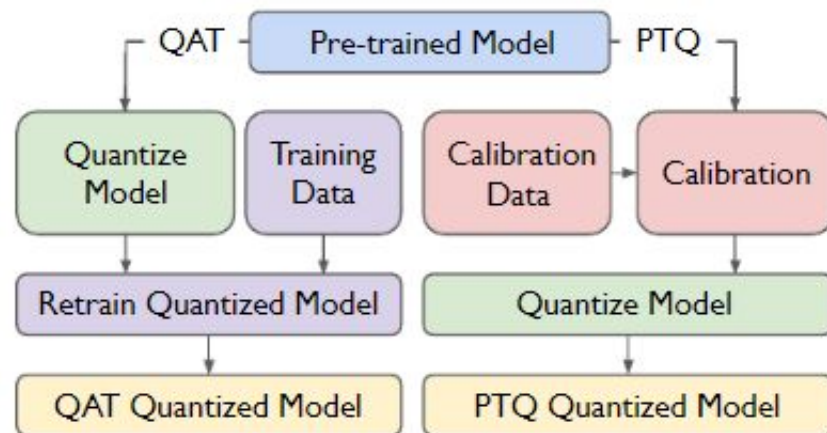
Data Management



4. Model & Parameter Optimization

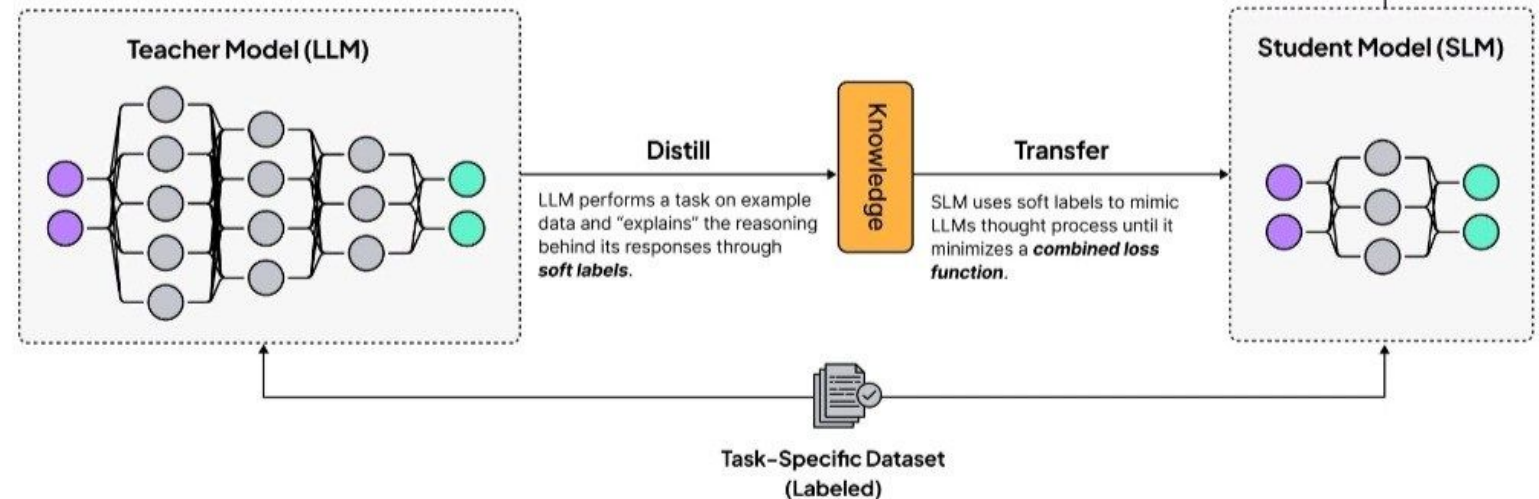
Model & Parameter Optimization

Model Quantization



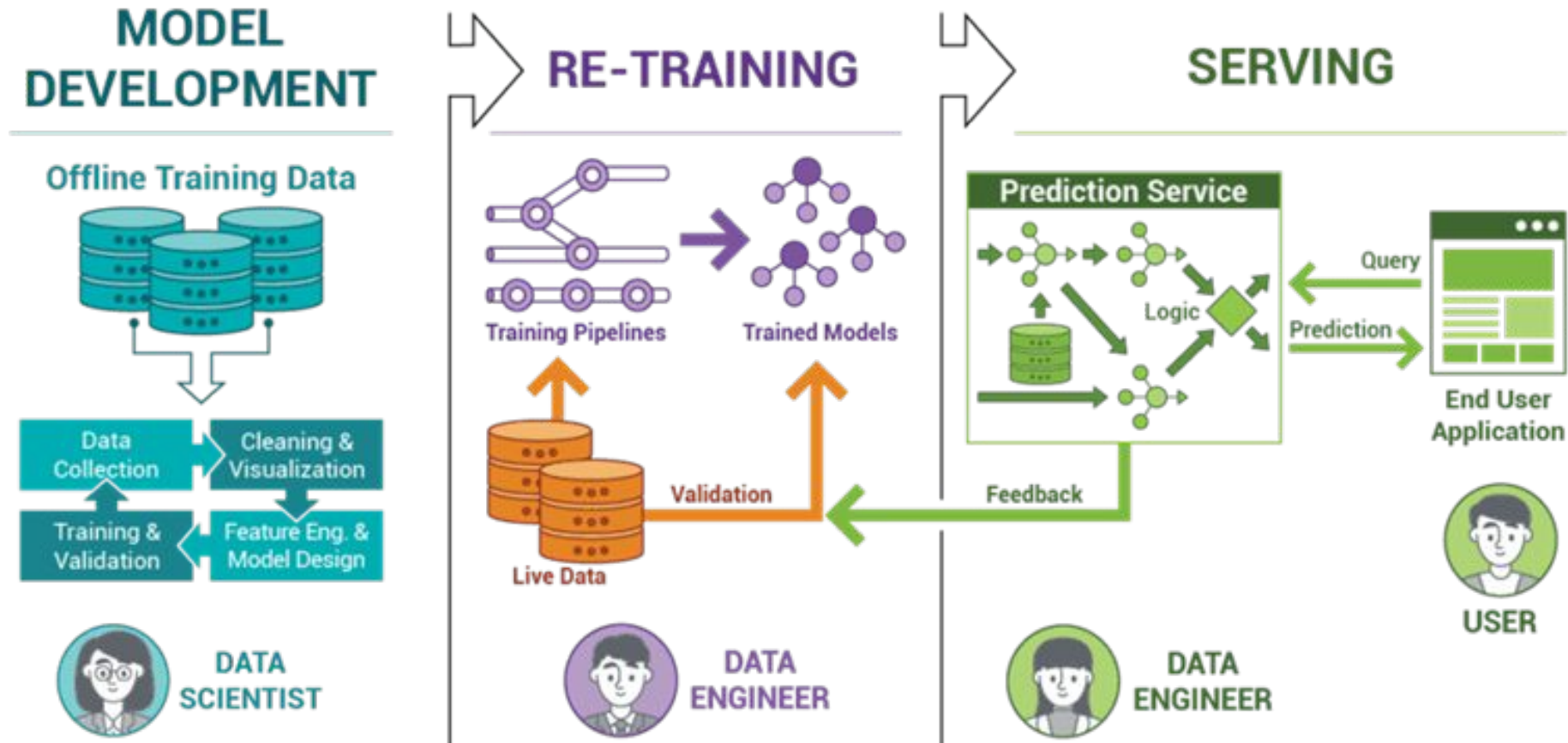
Model Distillation

transfers knowledge from a massive LLM to a SLM while maintaining (or improving) performance on a specific task.

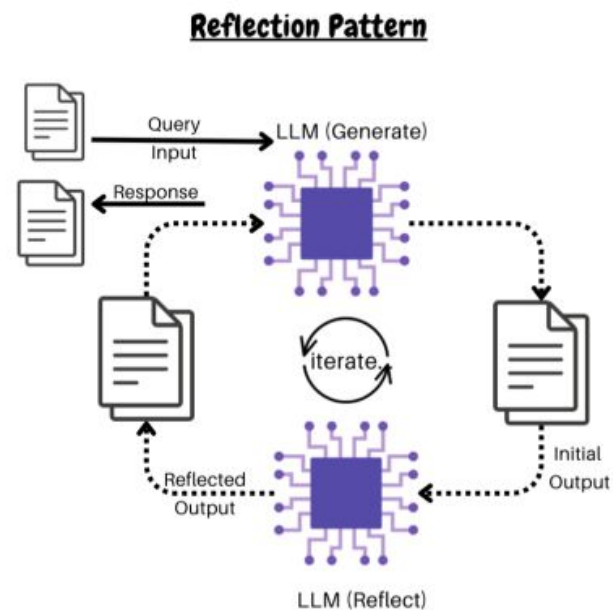


5. Deployment

Deployment

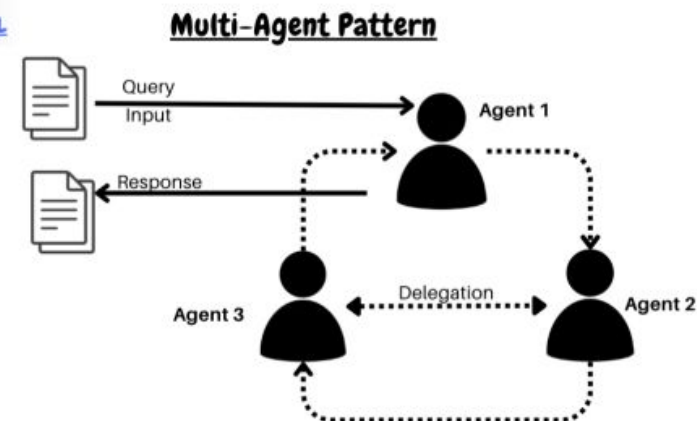
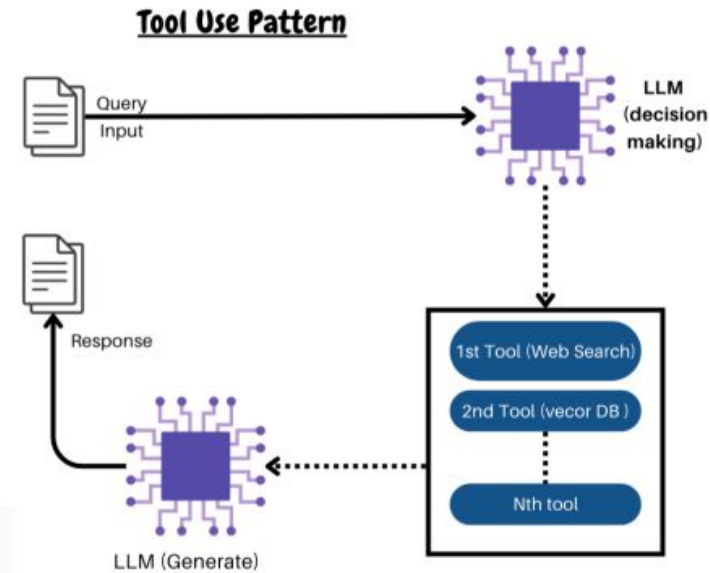
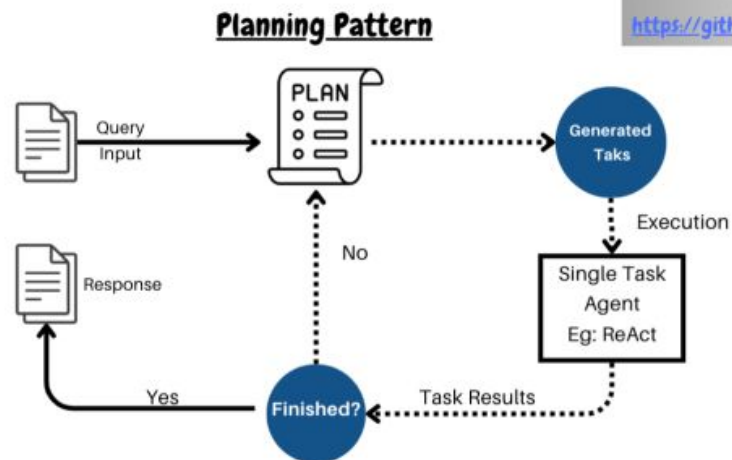


6. Agentic AI



Agentic AI Design Patterns

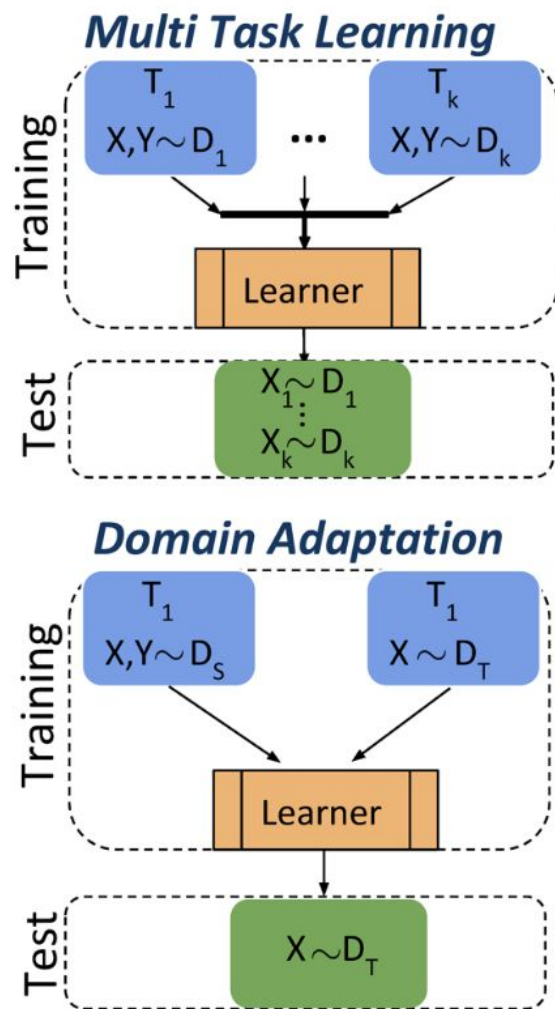
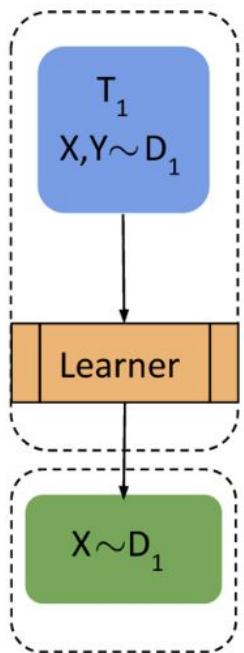
<https://github.com/OSsatyam/AI-ML>



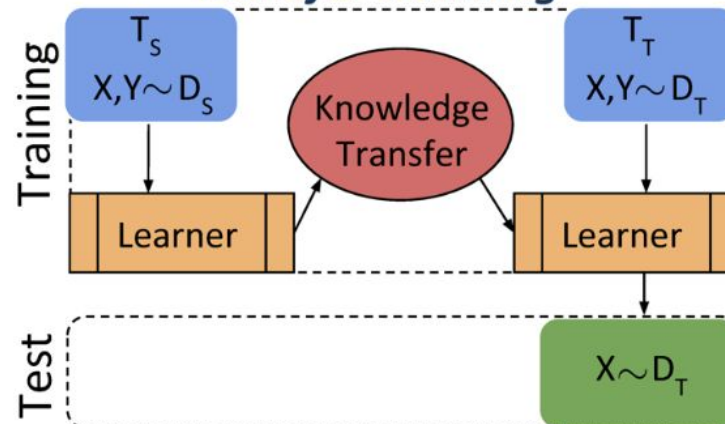
7. Continual Learning

Continual Learning

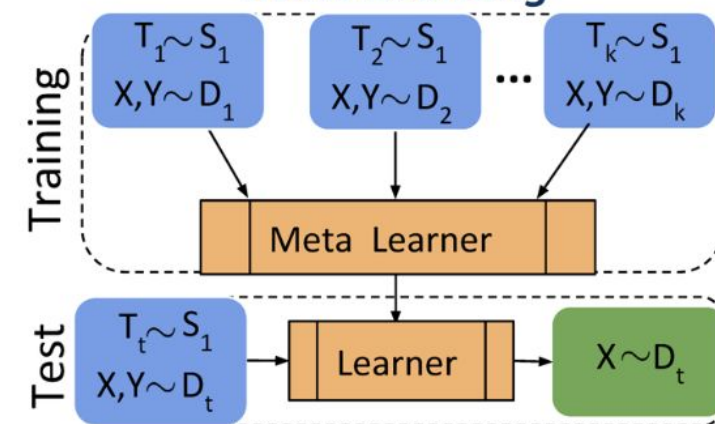
Standard Supervised Learning



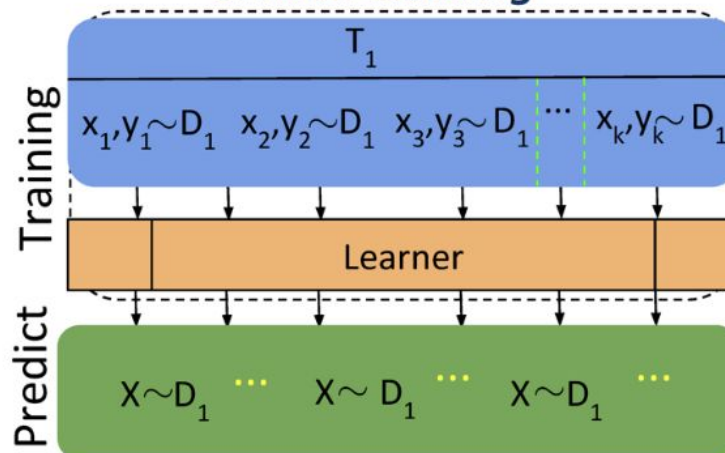
Transfer Learning



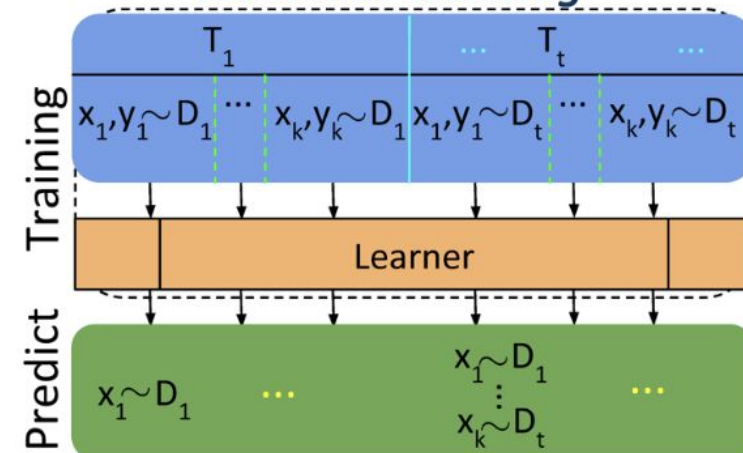
Meta Learning



Online Learning



Continual Learning



8. References

References

- Association for Computational Linguistics: <https://aclanthology.org/>
 - System Demonstration Track
 - Industry Track